Jinhao Li Ph.D. Candidate

lijinhao@sjtu.edu.cn
Research Homepage

Education

Shanghai Jiao Tong University, China 2021–Mar. 2027 *(Expected)*
Ph.D. in Mechanical Engineering Supervisors: Prof. Baoyang Lu and Prof. Guoying Gu

Hunan University, China 2017–2021
B.Eng. in Vehicle Engineering Advisor: Prof. Huigao Duan

Research Interests

- Iontronic meta-interfaces based on architected conducting polymers
- Soft wireless printed electronics for biomedical sensory systems

Research Experience

Cryogenic printing of three-dimensional hydrogel machines

- Proposed a multimaterial cryogenic printing technique for hydrogel meta-structures by harnessing a solvent phase transition strategy.
- Designed architected PEDOT:PSS-based interfaces to linearize sensing responses and suppress multimodal crosstalk in soft iontronic sensors.
- Developed untethered miniature robots and soft somatosensory prostheses like hydrogel aortic valves and dexterous neuroprosthetic hands.

Stretchable wireless bandage for deep infection monitoring

- Proposed a permeable epidermal multisensory system with hierarchical skin-mimetic architectures for monitoring thermomechanical dynamics.
- Developed multilayer printed processing circuits by engineering fabric-elastomer hybrids for stable soft-rigid electrical interconnection.
- Enabled early detection of periprosthetic joint infection within 4 hours to support timely intervention before antibiotic-tolerant biofilm formation and joint damage.

Publications and Patents

1. 3D-printed PEDOT:PSS for soft robotics.
   Jinhao Li[#], Jie Cao[#], Baoyang Lu*, Guoying Gu*.
   Nature Reviews Materials, 8, 604-622 (2023).
2. Multimaterial cryogenic printing of three-dimensional soft hydrogel machines.
   Jinhao Li, Jie Cao, Rong Bian, Rongtai Wan, Xiangyang Zhu*, Baoyang Lu*, Guoying Gu*.
   Nature Communications, 16, 185 (2025).
3. A wireless skin-mimetic bandage for continuous thermomechanical monitoring of peri-prosthetic joint infection.
   Jinhao Li[#], Jialian Xu[#], Yanzhe Zhao[#], Ting Xie, Dezhi Yang, Yuxuan Zhang, Lei Zhang, Dengfeng Li, Bing Yue, Xinhua Qu*, Guoying Gu*.
   Science Advances, Under review (2026).
4. Soft 3D lattice iontronic tactile sensor.
   Jinhao Li, Qinghua Yu, Zequn Shen, Dong Wang, Guoying Gu*.
   SmartBot, 1, e70002 (2025).
5. Large-scale zone-based approach to global modeling and optimization for a novel thermal management system of module-free lithium-ion battery.
   Jinhao Li, Fei Lei*, Wenhao Zhu, Zhifeng Qiu, Hailong Wu.
   Structural and Multidisciplinary Optimization, 64, 3621-3636 (2021).
6. Ultrathin and ultralight Zn micromesh-induced spatial-selection deposition for flexible high-specific-energy Zn-ion batteries.
   Huaizhi Liu[#], **Jinhao Li**[#], Xianan Zhang, Xiuxue Liu, Yu Yan, Fengjun Chen, Guanhua Zhang*, Huigao Duan*.
   Advanced Functional Materials, 31, 2106550 (2021).

7. Biomimetic honeycomb Zn anode enabled multi-field regulation toward highly stable flexible Zn-ion batteries.
   Huaizhi Liu, **Jinhao Li**, Donghai Wei, Xiuxue Liu, Zheren Cai, Hang Zhang, Zhisheng Lv, Lei Chen, Haicheng Li, Hongyu Luo, Yanli Zhao, Huihuang Yu, Xiaohu Wang, Fengjun Chen, Guanhua Zhang*, Huigao Duan*.
   Advanced Functional Materials, 33, 2300419 (2023).
8. Robust mechanically interlocked network ionogels.
   Mengling Yang, **Jinhao Li**, Chunyu Wang, Li Yang, Zhiwei Fan, Wenbin Wang, Guoquan Liu, Shaolei Qu, Zhaoming Zhang, Jiang Zou, Wei Yu, Guoying Gu*, Xuzhou Yan*.
   Angewandte Chemie International Edition, 64, e202423847 (2025).
9. High-stretchability, ultralow-hysteresis conducting polymer hydrogel strain sensors for soft machines.
   Zequn Shen, Zhilin Zhang, Ningbin Zhang, **Jinhao Li**, Peiwei Zhou, Faqi Hu, Yu Rong, Baoyang Lu*, Guoying Gu*.
   Advanced Materials, 34, 2203650 (2022).
10. Millimeter-scale fluid-driven soft robots.
    Rong Bian, Ningbin Zhang, Xinyu Yang, **Jinhao Li**, Dezhi Yang, Jieji Ren, Jiang Zou, Guoying Gu*.
    National Science Review, 12, nwaf413 (2025).
11. Three-dimensional Fabrication of hydrogels. China Patent, CN202411841590.0.
    Guoying Gu, **Jinhao Li**.
12. Fabrication of flexible patternable Zinc electrode. China Patent, CN202010532726.5.
    Guanhua Zhang, **Jinhao Li**, Huigao Duan, Jin Hu.

Grants and Awards

Principal Investigator, Young Scientists Fund for Doctoral Students (Grant No. 524B2047)
National Natural Science Foundation of China (NSFC) 2024-2026

Youth Talent Support Program for Doctoral Students
China Association for Science and Technology (CAST) 2025-2026

National Scholarship (Undergraduate and Ph.D.)
Ministry of Education of China 2025, 2018

Graduate Academic Rising Star (10 recipients annually)
The 10th Chinese Soft Robotics Conference 2025

Best Oral Presentation Award
IROS Workshop Tactile Sensing Toward Robot Dexterity and Intelligence 2025

Graduate Academic Rising Star Finalist (Top 1‰)
Shanghai Jiao Tong University 2025

Doctoral Graduate with Distinction
Shanghai Jiao Tong University 2026

Research Skills

Device development:
- Design: Structural modeling and optimization, Multimodal sensing, Iontronic interfaces
- Analysis: Multiphysics simulation, Kinetic modeling
- Polymer development: Conducting polymers, Hydrogels, Elastomer hybrids

Cleanroom microfabrication:
- Material growth: ALD, Sputtering, Thermal evaporation, Electro-/electroless-plating
- High-resolution patterning: Photolithography, Multimaterial 3D printing

System integration:
- Wearable electronics: Wireless electronic skin, Stretchable circuits, Flexible batteries
- Sensorized prostheses: Human-machine interaction, Sensory feedback loop